

SHOWRIN RAHMAN

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SUMMARY

Computer Vision and AI researcher with experience spanning real-world deployment and peer-reviewed research. Specialized in deep learning, image segmentation, diffusion models and remote sensing, with strong model development and experimental evaluation skills.

EDUCATION

- BRAC University** June 2021 - September 2025
Bachelor of Science in Computer Science, CGPA: 3.62 Dhaka, Bangladesh
- Bard College** January 2024 – May 2024
Exchange Semester New York, USA

RESEARCH & PUBLICATIONS

- Multi-Scale Feature Fusion with Adaptive Attention for Robust Image Deblurring** (*IEEE Eurocon 2025*): Multi-scale adaptive attention for spatially varying blur, improving PSNR/SSIM while reducing computational cost.
- TWIG: Two-Step Image Generation using Segmentation Masks in Diffusion Models** (*Springer IFIP 2025*): Segmentation-guided diffusion that separates structure and appearance to generate copyright-safe images.
- Riverbank Change Detection in the Jamuna River Basin, Bangladesh** (*IEEE ACDSA 2025*): Satellite-based machine learning analysis of riverbank erosion and accretion in a highly dynamic river system.
- MAMT: Masked Autoencoder and Mamba-Based Self-Supervised Segmentation of SAR Imagery for Riverbank Erosion Detection** (*Undergraduate Thesis*): Proposed MAMT, a self-supervised MAE–Mamba framework with GAN-based temporal gap reconstruction for SAR land–water segmentation under sparse annotations.

RESEARCH COLLABORATION

- Research Collaborator (R&D)** July 2025 – December 2025
276 Holdings Co., Ltd. (AI & Data Division) South Korea
Project: **AI Solution for Automatic Document Recognition (FlowPay)**
Role: **Independent Researcher, ML & DL Engineer**
- R&D Collaboration Certificate: [View Certificate](#)
Contributed to an industry RD project in Computer Vision and Intelligent Document Processing (IDP), developing OCR pipelines using EasyOCR and Tesseract with document-specific preprocessing, improving accuracy via MFSAA-based image deblurring (IEEE Eurocon 2025), and conducting systematic performance evaluation with experimental prototyping and weekly reporting.

EXPERIENCE

- ACI PLC — AI Application Developer** Feb 2026 – Present
Dhaka, Bangladesh
 - Built a **full-stack AI Mock Exam simulator** using **FastAPI** and **React (TypeScript + Tailwind)** with auth, dashboards, and analytics. Integrated **Gemini API (Mistral fallback)** with schema validation and caching for reliable generation. Developed AI-evaluated speaking, writing, and listening modules with timed exams, multi-criteria scoring, and PDF feedback, plus multi-accent listening via **Edge TTS (fallback: ElevenLabs)**. Deployed using **Docker** on production servers.
 - Optimized a **LangChain-based RAG system** (Gemini API + Google Drive search), reducing latency from **1 min to 30s** via improved retrieval, chunking, and orchestration. Integrated **Google Calendar API** and Drive-based file management with automatic chunk sync/delete. Enhanced chatbot orchestration by isolating context streams to reduce hallucinations and enable source-linked responses and PDF merging for internal team use. Deployed with **Docker** on server infrastructure.
 - Used **AI-assisted development tools** (Cursor, GitHub Copilot, Claude Code) to accelerate development, prototyping, and debugging workflows.
- Associate AI Engineer** October 2025 – February 2026
Ontik Technology Dhaka, Bangladesh
 - Built a **location-intelligence SaaS** for crime analytics and livability scoring using **FastAPI**, **PostgreSQL**, **Celery**, **Redis**, **n8n**-driven data pipelines, and **GCP** deployment.

- Developed an **AI chatbot** automating email ingestion and academic information retrieval using **IMAP**, **Pinecone** vector search, **CrewAI** agents, and **GCP**.
- Engineered a **FastAPI** and **LangChain** based AI platform with **MCP tooling** for property data automation and agent-driven workflows.
- Built **microservices** using **FastAPI**, deployed on **GCP Cloud Run** and **AWS EC2** with **Docker**, connected to **Cloud SQL** and **Pinecone** for scalable, low-latency AI services.
- Designed and optimized **multi-agent workflows**, **RAG pipelines**, and automation triggers, ensuring modularity, resilience, and observability across deployments.
- Followed **Agile sprints**, prepared **technical documentation**, and delivered project **presentations** to stakeholders.
- Tech Stack: **Python**, **CrewAI**, **LangChain**, **MCP Tools**, **Google ADK**, **GCP (Cloud Run, SQL, Scheduler)**, **AWS EC2**, **Pinecone**, **n8n**, **FastAPI**, **Docker**
- **AI Intern** **July 2025 – October 2025**
Ontik Technology Dhaka, Bangladesh
 - Contributed to internal **AI infrastructure** setup for model deployment, versioning, and workflow orchestration.
 - Integrated **n8n automation workflows** with **Google APIs (Drive, Sheets)** for real-time data synchronization.
 - Developed automation and agent workflows using **CrewAI**, **n8n**, and **Selenium** for internal process efficiency.
 - Participated in **Agile sprints**, documented workflows, and delivered internal **presentations** on project progress.
- **Coding Instructor (Part-Time, Remote)** **December 2024 – July 2025**
Dreamers Academy Dhaka, Bangladesh
 - Taught 100+ students beginner-level Python, JavaScript, HTML, and CSS.
- **Quality Assurance Analyst Intern (Remote)** **June 2024 – December 2024**
Sysonex Dhaka, Bangladesh
 - Led 60+ scrum meetings and 10+ development sprints, ensuring effective project coordination.
 - Reduced bug backlog by 15% through comprehensive test case development and Jira reporting.
 - Designed and executed 80+ test cases, uncovering and reporting bugs via comprehensive flow testing

SKILLS

- **Programming Languages:** Python, JavaScript, C
- **Machine Learning & Deep Learning:** TensorFlow, Keras, PyTorch, scikit-learn, NLTK, OpenCV, Scikit-LLM
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **LLMs & Generative AI:** Hugging Face, embeddings, RAG pipelines
- **Agentic AI & Automation:** CrewAI, n8n, Google ADK
- **Backend & Frontend Development:** FastAPI, Express.js, Node.js, React.js, REST APIs
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Cloud & MLOps:** AWS EC2, Google Cloud Platform (GCP), Docker, GitHub Actions
- **Web Scraping & Automation:** BeautifulSoup, Selenium
- **Tools & Platforms:** Git, GitHub, Linux (Ubuntu), Kaggle, Jira, Confluence, Roboflow
- **Basic Robotics:** Arduino

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **BRACU Mongol Tori — Senior I** **February 2022 – December 2023**
Student Organization Dhaka, Bangladesh
 - Built an offline rover mapping system with OSM achieving 20% efficiency gain; utilized OpenCV, Roboflow, Kaggle datasets, Git/GitHub, and developed the frontend for the BRACU Mongol Tori website.
- **Robotics Club of BRAC University (ROBU) — Secretary** **January 2024 – June 2024**
Student Organization Dhaka, Bangladesh
 - Served as a panelist in a 40-member Research and Project Management department, managed the flagship Basics of Robotics course with Arduino, and developed 5+ projects using ROS, OpenCV.

HONORS & ACHIEVEMENTS

- **OSUN Undergraduate Mobility Scholarship:** Awarded full funding (USD \$41,019) for Spring 2024 semester at Bard College, covering tuition, accommodation, living expenses, insurance, and travel
- **2nd Runner-Up:** 3rd Kibo Robot Programming Challenge 2022, organized by Japan Aerospace Exploration Agency (JAXA)
- **Top 5 Finalist:** FOREWARN Disaster Hackathon 2.0 (competed against 84 teams)
- **Top 15 Selection:** Banglalink Womentor Program (selected from 400 applicants)